

Arjuna JEE 2027

Maths by Amarnath Anand Sir

Quadratic Equations

DPP: 5

Total Duration - 43 Min

- Q1** If α is a roots of equation $4x^2 + 2x - 1 = 0$ and $f(x) = 4x^3 - 3x + 1$, then $2(f(\alpha) + \alpha)$ is:
 (A) 0 (B) -1
 (C) 1 (D) None of these
- Q2** The value of a for which the equation $(a^2 + 4a + 3)x^2 + (a^2 - a - 2)x + (a + 1)a = 0$ has more than two roots is-
 (A) 1 (B) 2
 (C) -2 (D) -1
- Q3** The value of $4 + \frac{1}{5 + \frac{1}{4 + \frac{1}{5 + \frac{1}{4 + \dots \infty}}}}$ is
 (A) $2 + \frac{2}{5}\sqrt{30}$
 (B) $2 + \frac{4}{\sqrt{5}}\sqrt{30}$
 (C) $5 + \frac{2}{5}\sqrt{30}$
 (D) $4 + \frac{4}{\sqrt{5}}\sqrt{30}$
- Q4** The number of solutions of the equation $\log_4(x - 1) = \log_2(x - 3)$ is ____.
- Q5** Difference between the corresponding roots of $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ is same and $a \neq b$, then
 (A) $a + b + 4 = 0$ (B) $a + b - 4 = 0$
 (C) $a - b - 4 = 0$ (D) $a - b + 4 = 0$
- Q6** If one root of the equation $x^2 + px + q = 0$ is the square of the other, then
 (A) $p^3 + q^2 - q(3p + 1) = 0$
 (B) $p^3 + q^2 + q(1 + 3p) = 0$
 (C) $p^3 + q^2 + q(3p - 1) = 0$
 (D) $p^3 + q^2 + q(1 - 3p) = 0$
- Q7** If the difference of the roots of $x^2 - px + q = 0$ is unity, then
 (A) $p^2 + 4q = 1$
 (B) $p^2 - 4q = 1$
 (C) $p^2 + 4q^2 = (1 + 2q)^2$
 (D) $4p^2 + q^2 = (1 + 2q)^2$
- Q8** The least value of k which makes the roots of the equation $x^2 + 5x + k = 0$ imaginary is
 (A) 4 (B) 5
 (C) 6 (D) 7
- Q9** If α and β are the roots of equation $x^2 - 7x + 1 = 0$ then find the value of $\frac{1}{(\alpha-7)^2} + \frac{1}{(\beta-7)^2}$?
- Q10** If α, β are the roots of the equation $ax^2 + bx + c = 0$, then equation whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$ is
 (A) $acx^2 + (a + c)bx + (a + c)^2 = 0$
 (B) $abx^2 + (a + c)bx + (a + c)^2 = 0$
 (C) $acx^2 + (a + b)cx + (a + c)^2 = 0$
 (D) $acx^3 + (a + c)bx + (a + c)^3 = 0$
- Q11** If the sum of the roots of the equation $ax^2 + bx + c = 0$ be equal to the sum of their squares, then
 (A) $a(a + b) = 2bc$
 (B) $c(a + c) = 2ab$
 (C) $b(a + b) = 2ac$
 (D) $b(a + b) = ac$
- Q12** Roots of the equation $x^2 + bx - c = 0$ ($b, c > 0$) are
 (A) Both positive
 (B) Both negative
 (C) Of opposite sign



(D) None of these

Q13 If α, β are roots of the equation $ax^2 + bx + c = 0$, then the value of $\alpha^3 + \beta^3$ is equal to

- (A) $\frac{3abc+b^3}{a^3}$
 (B) $\frac{a^3+b^3}{3abc}$
 (C) $\frac{3abc-b^3}{a^3}$
 (D) $\frac{-(3abc+b^3)}{a^3}$

Q14 If one real root of the quadratic equation $81x^2 + kx + 256 = 0$ is cube the other root, then a value of k is

- (A) -81 (B) 100
 (C) 144 (D) -300

Q15 The roots of the given equation $(p - q)x^2 + (q - r)x + (r - p) = 0$ are

- (A) $\frac{p-q}{r-p}, 1$
 (B) $\frac{q-r}{p-q}, 1$
 (C) $\frac{r-p}{p-q}, 1$
 (D) $1, \frac{q-r}{p-q}$

Q16 If one root of $5x^2 + 13x + k = 0$ is reciprocal of the other, then $k =$

- (A) 0 (B) 5
 (C) $1/6$ (D) 6

Q17 If $P(x) = ax^2 + bx + c$ and $Q(x) = -ax^2 + dx + c$, where $ac \neq 0$, then $P(x) \cdot Q(x) = 0$ has at least

- (A) Four real roots
 (B) Two real roots
 (C) Four imaginary roots
 (D) None of these

Q18 If the roots of the equations $px^2 + 2qx + r = 0$ and $qx^2 - 2\sqrt{pr}x + q = 0$ be real, then

- (A) $p = q$
 (B) $q^2 = pr$
 (C) $p^2 = qr$
 (D) $r^2 = pq$



Answer Key

Q1 (C)

Q2 (D)

Q3 (A)

Q4 1

Q5 (A)

Q6 (D)

Q7 (B, C)

Q8 (D)

Q9 47

Q10 (A)

Q11 (C)

Q12 (C)

Q13 (C)

Q14 (D)

Q15 (C)

Q16 (B)

Q17 (B)

Q18 (B)



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